Palindrome Partitioning Min Cuts

AlgoExpert · Very Hard · Dynamic Programming

The problem

- Fewest cuts so every piece is a palindrome.

Intuition

- Precompute which substrings are palindromes, then fewest cuts for each prefix.

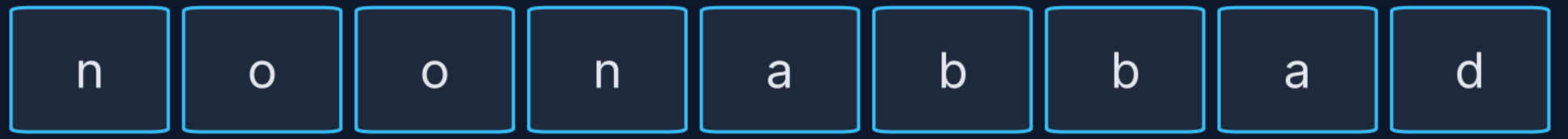
Brute force vs optimal

- Try all partitions → exponential
- Palindrome table + cuts DP → $O(n^2)$
- Expand-around-centre variant

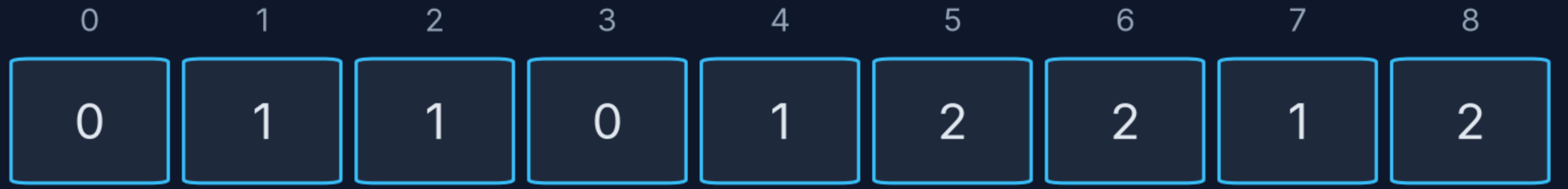
Approach

- 1. Palindrome table by expanding lengths
- 2. Cuts of i is zero if the prefix is a palindrome
- 3. Otherwise min over j of cuts plus one

Worked example



min cuts for prefix ending here



cuts[j] = min over palindromes s[i..j] of cuts[i−1] + 1

Worked example




```
noon | a | bbad? → 'noon' 'a' 'bb' 'a' 'd' … min cuts = 2
```

Answer: 2 cuts

Java solution

```java
public class PalindromePartitioningMinCuts {
    public static int palindromePartitioningMinCuts(String s) {
        int n = s.length();
        boolean[][] pal = new boolean[n][n];
        int[] cuts = new int[n];
        for (int j = 0; j < n; j++) {
            cuts[j] = j;
            for (int i = 0; i <= j; i++)
                if (s.charAt(i) == s.charAt(j) && (j - i < 2 || pal[i + 1][j - 1])) {
                    pal[i][j] = true;
                    cuts[j] = i == 0 ? 0 : Math.min(cuts[j], cuts[i - 1] + 1);
                }
        }
        return n == 0 ? 0 : cuts[n - 1];
    }
}
```

Complexity

- Time: O(n^2)
- Space: O(n^2)

Edge cases & pitfalls

- Whole string palindrome → 0
- Single chars always palindromes

Interview tips & follow-ups

- Mention centre-expansion O(1) space for pal

Recap

- Precompute which substrings are palindromes, then fewest cuts for each prefix.
- Complexity: O(n^2)